

Python Operators :

$x + y$
 $x - y$
 $x * y$

operators

Types of Operators :

1. Arithmetic Operators
2. Assignment Operators
3. Relational Operators
4. Logical Operators
5. Bitwise Operators ?
6. Identity Operators ?

1. Arithmetic Operators

$+$ → Addition

$-$ → Subtraction

$*$ → Multiplication

$/$ → division

$//$ → floor division

$\%$ → modulus operator

$**$ → Exponentiation *left most*

round off value → nearest round off value

$10/3 = 3$
 $10/3 = 3.33$

modulus($\%$):-

→ It gives remainder as result

$$3 \overline{)10}^3$$

Ex- $10 \% 3 = 1$

$10 \% 5 = 0$

Note:- $a \% b$

If $a < b$
 $\downarrow$
 $ans = a$

$2 \% 5 = 2$

$3 \overline{) 10}$

$- 9$

$\underline{1}$

$5 \overline{) 10}$

$- 10$

$\underline{0}$

Exponentiation :-

$2^2 = 4 \Rightarrow 2 \times 2$

$2^3 = 8 \Rightarrow 2 \times 2 \times 2$

$3^3 = 27 \Rightarrow 3 \times 3 \times 3$

$10^3 = 1000 \Rightarrow 10 \times 10 \times 10$

2. Assignment Operators $\rightarrow$ It is combination of arithmetic & equals operators

$+=$

$-=$

$\times=$

$/=$

$//=$

$\%=$

$\times\times=$

$x = 10$

$x += 1$

$print(x)$ $O/P \Rightarrow 11$

$x += 1 \rightarrow x = x + 1$

$x = 10 + 1 \Rightarrow 11$

$x = 10$

$x -= 5$

$x -= 5 \rightarrow x = x - 5$

$x = 10$

$x -= 5$

`print(x)`

$01p \Rightarrow 5$

$x -= 5 \rightarrow x = x - 5$

$x = 10 - 5 = 5$

3. Relational Operators

→ output of this either **TRUE OR FALSE**

$==$

$>$

$<$

$\geq$

$\leq$

$!=$

$x = 10 \quad y = 10 \quad x == y \rightarrow \text{True}$

$x = 2 \quad y = 10 \quad x == y \rightarrow \text{false}$

$a = 10 \quad b = 11$

$a > b \rightarrow 10 > 11 \times \text{false}$

$x = 10$

$y = 2$

$x > 2 \rightarrow x > 2 \text{ OR } x == 2$
↓
True

$!=$ → not equals

$x = 10 \quad y = 5$

$x != y \rightarrow \text{True} \checkmark$

$x = 10 \quad y = 10$

$x != y \rightarrow \text{false} \checkmark$

4. Logical Operators

AND

OR

both are true → then TRUE

$x > 2 \text{ and } x > 10$
 $x > 2 \text{ OR } x > 10$

↑

either one is true → then true